



VisionCore: Adapting Efficient Visual Answering Through Enhanced Spatial Reasoning

Ufuk Ozkul¹, Levent Karacan², and Cemil Zalluhoglu¹

¹ Department of Computer Engineering, Hacettepe University, Ankara, Turkey
ufukozkul@hacettepe.edu.tr cemil@cs.hacettepe.edu.tr

² Department of Computer Engineering, Gaziantep University, Gaziantep, Turkey
lkaracan@gantep.edu.tr

Abstract. Integrating spatially aware instruction tuning with Low-Rank Adaptation (LoRA) in pre-trained small-scale models unlocks enhanced performance in vision-language tasks. Although such models excel on general-domain data, they frequently lack the precise spatial reasoning and computational efficiency observed in large-scale architectures. Despite generating detailed textual descriptions, these models frequently struggle with tasks that require fundamental spatial reasoning abilities—such as distinguishing left from right or accurately discerning the relative positions of objects within a given context. This work investigates how integrating image-space coordinate annotations from the GQA dataset enhances spatial awareness in the Deep Seek 1.3B Visual Language Model (VLM). By freezing pre-trained weights and introducing specialized, trainable low-rank matrices through LoRA, only an additional 15 million parameters (1.15% of the overall model) and 60MB of memory are required, yielding a 16% performance improvement. Experimental evaluations reveal that this combined approach achieves an accuracy of 60.3%, surpassing comparable small-scale models and approaching the performance of large-scale architectures (63.5% accuracy) despite utilizing only 1.3B parameters—five times fewer than its larger counterparts. This efficiency-performance trade-off highlights the achievability of lightweight adaptation for the capability gap between small- and large-scale architectures in vision-language tasks.

1 Introduction

Human-like visual understanding in vision-language tasks requires spatial awareness—the ability to infer object positions, and relationships [18]. While large vision-language models (VLMs, e.g., 7B+ parameters) excel at spatial reasoning, their computational demands hinder real-world deployment [3]. Small-scale VLMs (1.3B parameters) address efficiency but suffer from critical limitations:

Limited Reasoning: Small-scale models exhibit deficiencies in spatial comprehension and structural analysis, hindering their ability to interpret intricate scene configurations. Challenges persist in tasks requiring relational reasoning, such as identifying object positions (e.g., left/right dependencies) or parsing multi-object interactions [20].

Limited Language Capabilities: Compact VLMs often misinterpret linguistic complexity and lack contextual depth, leading to semantically inconsistent or hallucinated outputs [25]. Deficiencies in semantic accuracy and discourse coherence further widen the performance gap with larger architectures [28].



Question: Is the fence made of cement or aluminum?

Prediction: The fence is made of aluminum. and scene description: The metal, chain-link fence located at: $x=0$, $y=23$, $w=377$, $h=380$ and in front of the tree that is located at: $x=193$, $y=112$, $w=70$, $h=105$. The small, green tree located at: $x=193$, $y=112$, $w=70$, $h=105$ and to the right of the pole...

Fig. 1. Example of spatial-material reasoning: The base model misclassifies the fence material and omits spatial details. After LoRA tuning, Vision Core corrects both errors, localizing the aluminum fence with coordinates and directional cues.

Recent advances in vision-language models (VLMs) such as MiniGPT-4 [29], Gemini 1.5 [22], and LocVLM-L [20] have demonstrated human-like understanding and response capabilities in linguistic tasks, nearly fulfilling the demands of the Visual Question Answering (VQA) field. However, these models continue to face notable constraints. For instance, when presented with an image of a kitchen scene containing a microwave to the left of a coffee maker and asked, ‘Is the microwave to the left or right of the coffee maker?’, MiniGPT-4 incorrectly answers, ‘The microwave is on the right side,’ while Gemini 1.5 generates, ‘The microwave is placed above the coffee maker.’ In contrast, LocVLM-L correctly identifies the spatial relationship: ‘The microwave is positioned to the left of the coffee maker.’ However, LocVLM-L requires $3\times$ more GPU memory (24GB vs. 8GB) for inference compared to MiniGPT-4, highlighting its computational inefficiency. These challenges underscore a broader issue in the design of large-scale language vision models: achieving a balance between computational efficiency

and description accuracy. Typically, larger models (with parameters exceeding 7 billion) can generate accurate visual answers and detailed image captions by effectively handling complex visual tasks, but this performance comes with increased demands for computational power and memory. In contrast, small-scale vision-language models offer reduced resource requirements and faster training speeds; however, their limited parameter capacity often results in suboptimal performance in spatial reasoning and precise visual description. Addressing this gap is crucial for advancing VLMs that are both efficient and capable of robust spatial understanding.

The dual challenges of language capability degradation and limited spatial reasoning in small-scale VLMs are addressed through parameter-efficient training (PEFT) with Low-Rank Adaptation (LoRA) [6]. The GQA dataset [7] is integrated to enhance standard annotations with spatial metadata—including bounding box coordinates (x, y, width, height) and directional relationships (e.g., left/right)—thereby injecting explicit spatial priors. LoRA optimizes this process by selectively updating low-rank matrices (1.15% of parameters, 15M/1.3B), preserving the base model’s pre-trained knowledge while reducing computational overhead. To further align visual and textual reasoning, structured prompts derived from GQA’s scene descriptions are introduced, encoding spatial directions (e.g., above/below) and object coordinates. This combined strategy achieves **60.3% accuracy** on the GQA benchmark—closing 72% of the performance gap with 7B, enabling robust spatial understanding with minimal resource requirements (2.5 days on 2×V100 GPUs) compared to full-scale retraining.

2 Literature Review

Prior Approaches in Image Captioning: Initial image captioning models utilized encoder decoder designs, employing Convolutional Neural Networks (CNNs) for the encoding part, and for the decoding phase, the common architecture was Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks [4]. These have constraints like information bottlenecks and insufficient detailed visual comprehension.

Localization in Vision Language Models: BLIP [11] and other foundation vision-language models (VLMs) have prompted substantial research of localization in pictures under supervision (bounding box) [12, 17, 21, 27] and under the reinforcement [9] technique. The recovery of explicit localization information from model representations has allowed for more resilient performance on numerous challenges [20, 26]. Recent advances in LLM-based generative prediction enable explicit location integration into language modalities, enhancing VLM performance [2, 5, 13] [15, 16, 22, 24].

Vision Language Models: The introduction of attention resulted in considerable improvements in image captioning. Many works; LLava [14], MetaVl [19], LLamaVid [12] presented attention-based approaches to dynamically focus on

important parts of images during caption development. However, these models frequently failed with spatial thinking, which is essential for understanding complicated scenarios. Ranasinghe et al. [20] addressed this gap by introducing coordinate-based embeddings and spatially aware training datasets. Their findings highlight that models like BLIP-2 and LLaVA fail to localize objects accurately, particularly in spatially dependent tasks. This underscores the importance of incorporating explicit spatial reasoning in VLMs. MiniGPT-4 uses techniques like a linear adapter layer and fine-tuning on chosen datasets to provide extensive visual discussion [29]. Similarly, DeepSeek-VL [16] incorporates cross-attention algorithms to align visual and textual information, thereby enhancing performance on standards like COCO. On the other hand, Combining LLMs with visual encoders presents issues such as multimodal alignment and computational inefficiency, especially when expanding to huge datasets [26]. Also, instruction adjusting has been revealed as an important approach for matching VLMs to human-like task guidelines. That method uses GPT-4 to create multimodal instruction datasets, allowing models to react successfully to various specific tasks [15]. VILA [13] highlights the relevance of employing interleaved datasets to promote modality alignment and retain text only capability of LLM. FERRET [5] addresses issues in fine-grained visual grounding and spatial thinking, with the latter reducing object hallucination via hybrid area representations yet constraints continue in dealing with abstract notions and processing efficiency. Furthermore, Low-Rank adaption (LoRa) [6] has become an effective fine-tuning approach, allowing for rapid adaption of huge pre-trained models with few parameter changes. LoRa’s implementation in vision-language activities provides scalability without losing quality [13].

Although models like MiniGPT-4, LLaVA, and Gemini 1.5 have advanced vision-language tasks, they fall short in fine-grained spatial reasoning due to coarse feature extraction and a heavy reliance on large-scale architectures. In contrast, the proposed approach departs from these methods by employing parameter efficient training and rank adaptation instead of full model retraining, and by leveraging a complex dataset that integrates spatial location data with relative object positions. Specifically, precise image-space coordinate instructions from the GQA dataset are injected into a compact Deep Seek 1.3B model, and by integrating Low-Rank Adaptation (LoRA), enhanced spatial reasoning is achieved with only a 1.15% increase in parameters—delivering robust visual question-answering performance while significantly reducing computational and memory requirements compared to large-scale architectures.

3 Methodology

While large-scale models like LLaVA [14], CogVLM [24], and QwenVL [2] achieve strong performance on standardized benchmarks such as VQAv2 (Table 4), their ability to articulate fine-grained spatial relationships remains limited. For instance, when queried about a scene containing “a flower vase in front of curtains and atop a dinner table next to plates,” these models often provide generic

descriptions (e.g., “The vase is on the table”) without specifying directional relationships (“in front of”) or precise object adjacency (“next to the plates”). This underscores a critical gap: while large models excel at answering what questions (e.g., object identification), they struggle with where and how questions requiring relational reasoning. To evaluate spatial nuances, results are reported on the VQAv2 validation set (Val), as test labels are restricted to blind evaluation servers—a common practice in VQA research [1, 23]. The proposed method addresses these limitations by explicitly encoding spatial metadata (e.g., bounding boxes, directional cues) through parameter-efficient adaptation. For example, Vision Core generates precise descriptions such as “The vase ($x = 120, y = 45$) is in front of the curtains and adjacent to the plates,” resolving ambiguities inherent in large-scale models while maintaining computational efficiency.

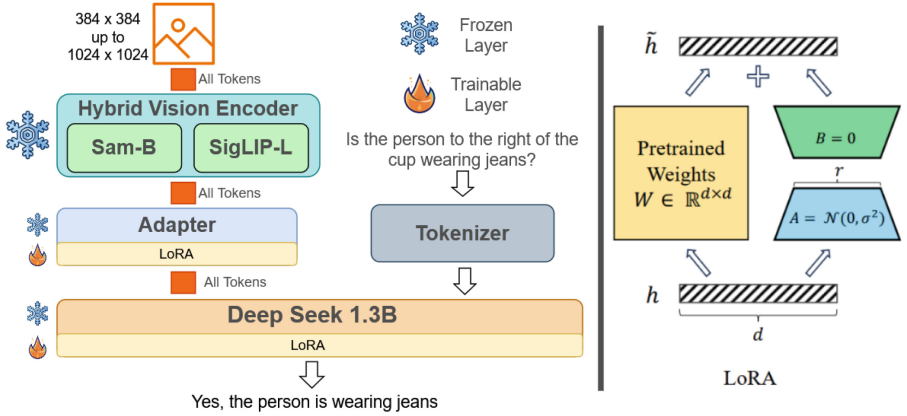


Fig. 2. Left: The overall architecture of the model where only the added parameters to the adapter layer and language model, are trainable. Right: the LoRA methodology [6]: standard LoRA using low-rank decomposition matrices. The only trained portions are A and B.

3.1 Model Architecture

The Vision Core model uses the architecture of the Deep Seek [16] VLM, consists of 3 parts namely; Vision Encoder, Adapter Layer, and Language Model.

Vision Encoder of the Deep Seek 1.3B uses the SigLIP [27] structure, vision-only self-supervised encoders, for improving the image processing capabilities that are required in the multi-modality vision tasks. This SigLIP encoder formed from the Sam-B [16] vision encoder and SigLIP-L where the Sam-B encoder processes low-level attributes while accepting high definition images (1024×1024) and SigLIP-L for the lower resolution images such as 384×384 . So this joint SigLIP vision encoder effectively encodes the images while preserving its high-quality features extracted from the input image. During the training,

the vision encoder part is frozen and it won't receive gradient update. In detail, the input image is rescaled to 1024×1024 format and the Sam-B part gives $64 \times 64 \times 256$ sized information. After that information appends to $96 \times 96 \times 256$ inside of the joint vision structure, then 2 layers of convolution operation resulting in $24 \times 24 \times 1024$ output which then resized to the 576×1024 final format by the Sam-B encoder. At the same time, the SigLIP-L also processes the input image to provide a 576×1024 low-resolution map which finally combined with the high-resolution output of the Sam-B (576×1024) and resulting 576×2048 attributes (576 tokens with R^{2048} dimension) for demonstrating the input image. In sum, the SigLIP encoder processes the input image to a 576×2048 format to effectively represent and maintain the high and low-resolution features where the number 576 symbolizes the tokens with the dimension of 2048. Then this eventual map is passed to the Projector (Adapter) layer.

Adapter Layer consists of a two-layer joint MLP (multi-layer perceptron) that acts like a bridge between the vision encoder and the language model. The first layer processes the high-resolution information and the second layer handles the low-resolution input retrieved from the vision encoder part, and those high and low-resolution features merge to the dimension. Then the fused features transferred to the language part with help of the another MLP. The hybrid projector consists of two parallel linear transformations followed by GELU activation, with the output concatenation formulated as:

$$\begin{aligned} \mathbf{h}_{\text{high}} &= \text{GELU}(\mathbf{W}_{\text{high}}\mathbf{x} + \mathbf{b}_{\text{high}}) \\ \mathbf{h}_{\text{low}} &= \text{GELU}(\mathbf{W}_{\text{low}}\mathbf{x} + \mathbf{b}_{\text{low}}) \end{aligned} \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^{d_{\text{input}}}$ is the input vector, $\mathbf{W}_{\{\text{high}, \text{low}\}} \in \mathbb{R}^{(d_{\text{embed}}/2) \times d_{\text{input}}}$

Language Model is Deep Seek LLM which has a similar architecture as the Llama model. Explicitly the Deep Seek 1.3B VL model uses the Deep Seek 1.3B model which is trained with corpus size 500B. For the processing part, the input text where the text and the image place-holder token are combined and tokenized before fused with the image features.

3.2 Fine Tuning

Experimental results demonstrate that concise prompts outperform verbose instructions. For example, on the GQA test-dev split (Table 2), the 'brief answer' prompt achieves an overall accuracy of 44.3%, outperforming detailed prompts like 'Analyze object attributes...' (40.3%) and 'Describe key attributes...' (39.8%). Notably, the 'brief' prompt improves performance in compare-type questions by 9.6% (from 30.6% to 40.2%) and verify-type questions by 7.2% (from 50.1% to 57.3%). Conversely, overly complex prompts reduce accuracy in logical reasoning tasks by up to 8.2%, suggesting that brevity aligns better with the model's capacity for spatial reasoning. To enhance complex visual reasoning and address challenges associated with query-type questions, additional parameters are injected into selected layers. Given that full fine-tuning is computationally prohibitive for small-scale models, a parameter-efficient fine-tuning (PEFT) strategy—specifically Low-Rank Adaptation (LoRA)—is employed.

LoRA updates only the trainable low-rank decomposition matrices (denoted as A and B in Fig. 2) within transformer layers while freezing all other parameters. In this implementation, LoRA is applied across seven layers: four within the language model (i.e., “q_proj”, “k_proj”, “v_proj”, “o_proj”) and three within the adapter layer (i.e., “gate_proj”, “up_proj”, “down_proj”). These modifications introduce an additional 15 million parameters (only 1.15% of the total), while the inclusion of the hyperparameter “Rank” (r) controls the dimensionality of the decomposition matrices. This approach accelerates fine-tuning and efficiently adapts the model for complex spatial reasoning tasks without incurring the high computational costs of full-scale training.

For dataset preparation and tokenization, the GQA dataset is utilized along with its detailed scene descriptions, which provide bounding box information and relative spatial directions for objects. The questions and ground truth answers are merged with these scene descriptions; however, this combined input often exceeds the context length limit. To address this, the scene description is truncated to 1024 characters, resulting in a training text length of approximately 1150–1250 characters per entry.

4 Experiments

4.1 Baseline and Instruction Tuning

To understand the base model capacity and reasoning level, several experiments have been made. The initial experiments evaluate how prompting strategies influence model performance, adopting methodologies from prior work [8] on task-specific instruction tuning. Then model’s chain of thought capacity is measured based on the iterative thinking methodology [26]. The default system prompt (You are a helpful language and vision assistant. You can understand the visual content that the user provides, and assist the user with a variety of tasks using natural language) has been used and it merged with the input text and used in all the experiments. Baseline evaluations utilize the GQA test-dev split, comprising 12,578 spatially oriented visual questions categorized into five types: choose, compare, logical, query, and verify.

The corresponding details are; the choose type has 1129 questions, the compare type has 589 questions, the logical type has 1803 questions, the query type has 6805 questions and the verify type has 2252 questions. For example, the choose type questions with two possibilities, such as “Is it white or black?”, the compare type for comparison such as “Which animal is taller?”, the logical type involves reasoning/inferences such as “Are there more apples than oranges on the plate?”. The query type involves all open questions like “What color is the car left of the tree?” and lastly the verify type for yes/no questions. Training is conducted using an 8-bit quantized model with LoRA parameters set to Rank 16, LoRA Alpha 32, a learning rate of $2e-4$, and the AdamW optimizer. The initial baseline tables are below.

In light of the result of Table 1 model’s hyperparameters have been fixed to false since the desired format of the experiments is zero shot where the model

Table 1. Results of Changing Hyper-Parameters of Model with GQA Test-Dev Split

Use Cache	Use Sampling	Accuracy (%)
Yes	No	41.9
Yes	Yes	34.9
No	No	41.9

faces the inputs the first time, to enable that use the cache parameter to set the false. For that reason the experiments have decided to conduct on the model’s parameters do sample and use cache where both of them are false due to resulting higher accuracy on the GQA test dev split.

Table 2. Combined Results with Integrated Punctuation Corrections for Prompting and Iterative Thinking Experiments on GQA Test-Dev Split

Prompt/Iterative Thinking Prompt	Overall Score	Structural Accuracy				
	(%)	Chs	Cmp	Log	Qry	Vrf
Prompting Experiment						
No Prompt	41.9	65.5	30.6	40.8	36.5	50.1
Provide brief answer (No punct. correct)	44.2	63.7	40.2	49.4	35.6	57.3
Provide brief answer (With punct. correct)	44.3↑	63.7	40.2	49.4	35.8↑	57.3
Analyze attributes, materials, contrast	40.3	63.2	27.7	32.6	38.9	42.3
Describe attributes, analyze relevance	39.8	64.0	23.4	32.7	38.3	42.2
Iterative Thinking						
Review and provide detailed answer	31.1	55.5	22.0	18.4	35.2	19.1
Focus on size, color, texture, position	22.1	69.4	20.4	14.8	18.2	16.5
Focus details, brief answer, review	35.1	62.4	23.8	27.5	35.2	30.2
Final Model Results						
Final Vision Core	60.3	78.7	53.8	68.4	50.1	76.9

The results have been listed with only 1 full forward pass in both the prompting experiments (2 full forward in the iterative thinking experiment). Table 2 evaluates the untuned DeepSeek 1.3B model’s spatial reasoning capabilities, demonstrating why architectural enhancements (via LoRA and spatial instruction tuning) are critical for closing the performance gap with large-scale models. The “Provide a brief answer to that question” prompt has the best overall accuracy (44.2%) and improved compare type score (40.2%), exceeding both the no-prompt baseline (41.9%) and other detailed prompts. On the other hand, the detailed prompts (fourth and fifth rows) have surpassed the baseline and the best accuracy prompts in the context of Query type questions (~3% improvement)

however they outdistanced by the baseline and the best prompt in compare, logical and verify type questions. So this demonstrates that overly detailed prompts block the model’s capability. Also, the choose type is negatively affected by 2% however it stays constant around 63–65% indicating robustness against the prompting strategy, in contrast prompting technique has a significant effect on the other types. The brief answer prompt has heavily (7–10%) increased the scores of the compare, logical, and verify types where the most cases the expected answer is short besides that, it fails to reach the complex prompt scores in the query type which requires detailed reasoning to answer correctly, and the complex prompts satisfies that condition. In sum, the prompting technique faces the problem of Simplicity and Specificity trade-off. The brief prompt’s performance demonstrates that less detailed instructions can succeed better at the questions that need less attention. In contrast, too detailed instruction may misdirect and confuse the model. Moreover, performance varies dramatically by question type (+9% gain for Compare type with the brief answer prompt vs. -3% for Query type). This underlines that task-specific prompt is a must for systems that are designed for specialized areas like spatial awareness. Lastly, the query type becomes a challenge since it cannot even reach the 40% and has the lowest score in the entire experiment.

In the iterative thinking experiment conducted on the 2-iteration thinking method where the model generates the answer in the first iteration with the highest score prompt in the prompting experiment and then the generated answer is combined with the iterative thinking prompt listed in Table 2 and passed to the model to generate a new answer. Due to the nature of the 2-step thinking the time needed for conducting the experiments doubles compared to the prompt experiment. The goal of this experiment is to adapt the model to complex reasoning tasks while preserving its capability in simple questions.

The third prompt “Focus on the question details. Try to provide a brief answer. You can review your answer” has reached the highest score (35.1%) among the other iterative prompts even though these scores are still far behind the scores listed in the prompting experiment. The second prompt in the iterative thinking experiment which focuses on the image details including size, color, and texture, has surpassed the choose type questions and reached a 69.4% score. This shows that visual details tend to help in object selection and discrimination tasks where it is required in the choose type questions. The second prompt reached the highest choose score among the experiments but achieved the lowest overall accuracy. This case once again proves that the overly complex instructions distract the model in the context of task-specific reasoning. In conclusion of these experiments, the observed situation is the “Choose” type tasks that require visual indicators, whereas Logical/Verify tasks need deeper, context-aware thinking. Even the highest-performing prompt (35.1%) has great potential for improvement, highlighting the need for more advanced reasoning techniques.

By the result of all the experiments, the brief prompt derived from the prompt experiment has marked to apply training using it and the training mainly focuses

on improving its lowest score “query” type which forms the majority (54.1%) of GQA test dev split. The main reason is that it achieves the highest overall accuracy and it has room for improving the query type score (35.6%). Also, the other dataset VQAv2 [1] offers a punctuation correction method to efficiently measure the results of the model. The punctuation correction converts all the numbers into letters and eliminates the punctuation marks from the answer to prevent any mismatch between the predicted answer and the ground truth answer. When the punctuation method was applied to the highest accuracy score “brief” prompt, a slight improvement (0.2%) happened in the query type questions and a tiny improvement (0.1%) in overall accuracy where the original GQA evaluation method confused at the letter version of numbers. The punctuation method wasn’t applied to previous results and the “brief” prompt was selected to use in the training part as a baseline. Also, the baseline scores 53.2% in the VQAv2 Val dataset [1] in zero-shot evaluation.

4.2 Ablation Study

To demonstrate the impact of LoRA-based spatial instruction tuning, consider a test case where the model is queried: “Is the fence made of cement or aluminum? Then describe the scene objects by their spatial information.” The untrained DeepSeek 1.3B VLM incorrectly responds: “The fence is made of cement.” Notably, it omits spatial descriptions of scene objects entirely, reflecting its inability to jointly reason about material properties and object localization. After LoRA fine-tuning with spatial coordinate instructions, the model answers (Fig. 1): “The fence is made of aluminum and the metal, chain-link fence located at $x = 0$, $y = 23$, $w = 377$, $h = 380$ and in front of the tree which located...” This example highlights two critical improvements:

Correct Material Identification: The model now distinguishes fine-grained material attributes (cement vs. aluminum), reducing hallucination.

Spatial Precision: Quantitative coordinate references (e.g., “ $x = 0$, $y = 23$, $w = 377$, $h = 380$ ”) and directional relationships (e.g., “in front of”) demonstrate learned spatial reasoning, aligning with the GQA dataset’s explicit positional annotations. These qualitative results corroborate quantitative improvements in Table 2, where the trained model achieves a **16% accuracy gain** on query-type questions requiring joint material and spatial reasoning. The base model’s failure to localize objects or infer materials underscores the limitations of vanilla VLMs in fine-grained visual scene understanding, while Vision Core’s performance illustrates the efficacy of low-rank spatial adaptation.

The results have been presented to show how to additional 1.15% parameters affect the accuracy and handle the “query” type question challenge and LoRA technique for addressing the limited computational power in the training phase where LoRA parameters are only needed 60 MB (59.18) memory to store.

Table 3. Trained Lora Results with Different Dataset Sizes to Detect Catastrophic Forgetting using the GQA Test-Dev Split

Dataset Size	Halo Quest [25]	Overall Accuracy	Structural Type Accuracy				
			Choose	Compare	Logical	Query	Verify
10K	No	57.7	78.4	48.2	65.7	47.3	75
50K	No	58.9	76.7	53.3	67.3	48.2	77.3
50K	Yes	55.6	73.3	58.1	66.1	44.6	71
100K	No	60.3	78.7	53.8	68.4	50.1	76.9
100K	Yes	57.2	77.7	54.3	66	46.7	72.7
132K	No	59.4	75.7	58.4	68.2	48.6	76.9
132K	Yes	56.6	73.5	56	65.6	46	73.2

The parameter size of the Vision Core which uses the Deep Seek-VL 1.3B as a backbone, is compact. Since the language model is small-scaled and combined with the lightweight LoRA parameters at the last layer in the architecture is vulnerable to catastrophic forgetting, which weakens its ability to remember previously acquired knowledge in the fine-tuning process. The ratios are changing across different models and it is not always clear to detect. For example, in the LocVLM model, the authors decide to use the 200K dataset for the 7B model. For detecting the catastrophic forgetting and finding the suitable dataset size for the small-scale Vision Core model the catastrophic forgetting experiment has been conducted. As seen in Table 3, the overall accuracy improves when the dataset size increases up to 100K (57% $\rightarrow$ 60%). However, after that, the accuracy drops (0.9%) in the 132K dataset showing that the model has been affected by the forgetting. The performance degradation at 132K samples (vs. 100K) highlights a critical limitation of small-scale models (1.3B parameters): limited capacity to scale with dataset size. Unlike larger architectures (e.g., 7B+). Also, the method “negative prediction” introduced in LocVLM claims that putting the instructions that ask questions about the non-existent objects in the image will improve the performance of them odel [20]. By inspiring from that explanation, an additional dataset Halo Quest [25] which consists of 7K questions about non-existing objects, has combined with the previously prepared dataset resulting additional 7K entries and scores have been listed in Table 3. The trained Halo models change their answer structure from “chair is short” to “chair is not tall” illustrating that models have learned the negative prediction. Despite this models trained with additional Halo Quest data underperform in the context of overall accuracy. This indicates that Halo Quest may produce noise that negatively impacts generalizability. On the other hand, Halo Quest enhances the “Compare” accuracy, which can help models understand comparative distinctions but at the price of overall performance. Furthermore, the “query” challenge result has improved (35.8% to 50.1%) indicating that additional LoRA parameters have increased the reasoning capability of the model.

Table 4. Comparison of Models by Scale based on GQA Test-Dev and VQAv2 Val Accuracy Results

Small-Scale Models (LLM Parameter Size <7B)			
Model	LLM Parameter Size	GQA Test-Dev	VQAv2-Val
MetaVL [19]	375M	32	33.1
FEWVLMlarge [8]	740M	35.7	51.1
UnifiedQAv2 [23]	738M	38.4	57.5
VIT-L OPT1.3B OpenFlamingo [10]	2B	39.1	61.1
UnifiedQAv2 [23]	2.9B	42.3	62.1
VIT-G InstructBLIP OPT1.3B [10]	2.4B	43.6	70.7
ViT-L BLIP-2 FlanT5XL [11]	3.4B	44.4	62.6
VIT-G InstructBLIP FlanT5XL [10]	4.2B	48.6	72.6
Vision Core (proposed)	1.3B (+15M LoRA)	60.3	63.3
Large-Scale Models (LLM Parameter Size ≥7B)			
Model	LLM Parameter Size	GQA Test-Dev	VQAv2-Val
MiniGPT-4 [29]	13B	33.4	–
MiniGPT-4 (Finetune Vicuna) [29]	13B	43.5	–
HYDRA [9]	–	47.9	–
Qwen-VL [28]	7B	59.3	78.8
LLaVA-v1.5 [20]	7B	62	78.1
LocVLM-L [20]	7B	63.5	78.2

The Vision Core model, using the Deep Seek-VL 1.3B VLM backbone with 15M LoRA parameters, was evaluated on GQA and VQAv2. On GQA Test-Dev, it achieves 60.3%, outperforming larger models (MiniGPT-4 at 43.5% and UnifiedQAv2-2.9B at 42.3%) and approaching state-of-the-art 7B models (LLaVA-v1.5 at 62% and LocVLM-L at 63.5%). However, on VQAv2-Val it reaches 63.3%, trailing models like Qwen-VL (78.8%), which suggests a trade-off between parameter efficiency and generalization.

5 Conclusion

This paper introduces the Vision Core model to address two critical challenges in computational efficiency and limited spatial reasoning. By leveraging the small-scale DeepSeek 1.3B backbone and integrating LoRA-based parameter-efficient training, the model effectively reduces the computational burden while enhancing spatial understanding. The approach involves freezing the pre-trained model’s layers, except for the 15M LoRA parameters (1.15% of the model), which require only 60MB of additional memory. Furthermore, the incorporation of spatial scene descriptions derived from the GQA dataset enables the model to develop a more holistic understanding of object localization and spatial relationships (e.g., left, right, above, and below).

As a result, Vision Core achieves 60.3% accuracy on the GQA dataset, outperforming other small-scale models and approaching the performance of 7B-scale models such as LocVLM-L (63.5%), despite using $5\times$ fewer parameters. This study demonstrates that combining compact architectures, parameter-efficient fine-tuning, and spatial scene augmentation can yield competitive results in visual question-answering tasks, paving the way for more accessible and scalable vision-language solutions, particularly in resource-constrained environments.

5.1 Limitations

The limitation of the proposed Vision Core, utilizes LoRA-based fine-tuning, is its reduced flexibility in handling diverse input and output instructions. This issue arises because the additional parameters, located at the end of the transformer layers, are fine-tuned using a single conversation format. Consequently, the model becomes overly specialized to that specific wording and syntax, diminishing its ability to generalize to alternative or unfamiliar instruction styles. Such overfitting to a particular instruction pattern may undermine the model's robustness in real-world applications where input variability is common.

5.2 Future Work

Future research will focus on expanding the model's adaptability to diverse instruction styles by incorporating multi-style prompt augmentation during training, which may help mitigate overfitting to a single prompt format. Also, exploring adaptive fine-tuning strategies that dynamically adjust parameter updates based on task complexity could further enhance generalization and performance.

References

1. Agrawal, A., et al.: VQA: Visual question answering (2016). <https://arxiv.org/abs/1505.00468>
2. Bai, J., et al.: Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond (2023). <https://arxiv.org/abs/2308.12966>
3. Chen, K., et al.: Shikra: Unleashing multimodal LLM's referential dialogue magic (2023). <https://arxiv.org/abs/2306.15195>
4. Ghandi, T., Pourreza, H., Mahyar, H.: Deep learning approaches on image captioning: a review. *ACM Comput. Surv.* **56**(3), 1–39 (2023). <http://dx.doi.org/10.1145/3617592>
5. Ghosh, A., et al.: Exploring the frontier of vision-language models: A survey of current methodologies and future directions (2024). <https://arxiv.org/abs/2404.07214>
6. Hu, E.J., et al.: LoRA: Low-rank adaptation of large language models (2021). <https://arxiv.org/abs/2106.09685>
7. Hudson, D.A., Manning, C.D.: GQA: A new dataset for real-world visual reasoning and compositional question answering (2019). <https://arxiv.org/abs/1902.09506>

8. Jin, W., et al.: A good prompt is worth millions of parameters: Low-resource prompt-based learning for vision-language models (2022). <https://arxiv.org/abs/2110.08484>
9. Ke, F., et al.: HYDRA: A hyper agent for dynamic compositional visual reasoning (2024). <https://arxiv.org/abs/2403.12884>
10. Lee, J., et al.: Visual question answering instruction: Unlocking multimodal large language model to domain-specific visual multitasks (2024). <https://arxiv.org/abs/2402.08360>
11. Li, J., et al.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models (2023). <https://arxiv.org/abs/2301.12597>
12. Li, Y., Wang, C., Jia, J.: LLaMA-VID: An image is worth 2 tokens in large language models (2023). <https://arxiv.org/abs/2311.17043>
13. Lin, J., et al.: VILA: On pre-training for visual language models (2024). <https://arxiv.org/abs/2312.07533>
14. Liu, H., Li, C., Li, Y., Lee, Y.J.: Improved baselines with visual instruction tuning (2024). <https://arxiv.org/abs/2310.03744>
15. Liu, H., et al.: Visual instruction tuning (2023). <https://arxiv.org/abs/2304.08485>
16. Lu, H., et al.: DeepSeek-VL: Towards real-world vision-language understanding (2024). <https://arxiv.org/abs/2403.05525>
17. Luo, M., et al.: ‘just because you are right, doesn’t mean i am wrong’: Overcoming a bottleneck in the development and evaluation of open-ended visual question answering (VQA) tasks (2022). <https://arxiv.org/abs/2103.15022>
18. Marr, D.: Vision: A Computational Investigation into the Human Representation and Processing of Visual Information. The MIT Press (2010)
19. Monajatipoor, M., et al.: MetaVL: Transferring in-context learning ability from language models to vision-language models (2023). <https://arxiv.org/abs/2306.01311>
20. Ranasinghe, K., et al.: Learning to localize objects improves spatial reasoning in visual-LLMs (2024). <https://arxiv.org/abs/2404.07449>
21. Shen, Y., Xu, Z., Wang, Q., Cheng, Y., Yin, W., Huang, L.: Multimodal instruction tuning with conditional mixture of LoRA (2024). <https://arxiv.org/abs/2402.15896>
22. Team, G., Georgiev, P., et al.: Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context (2024). <https://arxiv.org/abs/2403.05530>
23. Tiong, A.M.H., et al.: Plug-and-play VQA: Zero-shot VQA by conjoining large pretrained models with zero training (2023). <https://arxiv.org/abs/2210.08773>
24. Wang, W., et al.: CogVLM: Visual expert for pretrained language models (2024). <https://arxiv.org/abs/2311.03079>
25. Wang, Z., et al.: HaloQuest: A visual hallucination dataset for advancing multimodal reasoning (2024). <https://arxiv.org/abs/2407.15680>
26. Wu, P., Xie, S.: V*: Guided visual search as a core mechanism in multimodal LLMs (2023). <https://arxiv.org/abs/2312.14135>
27. Zhai, X., Mustafa, B., Kolesnikov, A., Beyer, L.: Sigmoid loss for language image pre-training (2023). <https://arxiv.org/abs/2303.15343>
28. Zhang, D., et al.: MM-LLMs: Recent advances in multimodal large language models (2024). <https://arxiv.org/abs/2401.13601>
29. Zhu, D., et al.: MiniGPT-4: Enhancing vision-language understanding with advanced large language models (2023). <https://arxiv.org/abs/2304.10592>